

REACT+TYPESCRIPT DAY 1

Typing React Props

- What are props? (Contract between parent and child).
- How to define them (interface vs type alias - mention both, explain when to use which).
- Required vs Optional props (use of ?).
- Common prop types: Primitive, arrays, objects, functions (callbacks).
- Special props: children (React.ReactNode), className / style.
- Generic props (like for list components).
- Default props handling and type safety.

Typing useState

- How TypeScript infers types.
- When inference works (primitives).
- When you MUST use generics (null, undefined, objects, arrays, union types).
- Common pitfalls: null initial value and type narrowing.
- Typing state with complex objects or arrays.
- Typing update functions (functional updates).
- State with discriminated unions.

Typing Events (The Big 3 - Click, Change, Submit)

- Why use React's SyntheticEvent types instead of DOM native.
- MouseEvent: what it is, when to use currentTarget vs target.
- ChangeEvent: primarily for inputs, selects, textareas.
- FormEvent: for form submission, preventing default behavior.
- The naming convention: React.MouseEvent<HTMLButtonElement> etc.
- Explain the <T> generic for the specific HTML element.
- Explain when you don't need to type it (if using inline arrow functions with inference, but explain that explicit is safer).
- Other events: KeyboardEvent, FocusEvent.